## Evaluation of Responses to Fundamental Concepts

We use these questions (check the Prompts and Response.pdf) to assess each LLM's familiarity with the architecture, features, and design paradigms of the two selected SWSs. We submit the questions to the models before issuing any workflow generation prompts so that the evaluation can first examine whether the LLMs understand the underlying systems and align their reasoning with the experimental setting.

We pose all background questions to *GPT-4o, Gemini 2.5 Flash, and DeepSeek-V3*, and instruct each model to provide concise but informative answers. Two authors with substantial domain expertise in scientific workflows (with over five and nine years of experience, respectively) independently review the responses. We then cross-check the answers against authoritative literature and official documentation to verify their accuracy and relevance.

**Assessment of LLM Understanding of Core Workflow Concepts**: We find that all three models produce technically accurate responses that align with the intended meaning of each question. To assess their effectiveness more systematically, we evaluate the responses using six qualitative criteria: *correctness, clarity, completeness, terminology, structure, and usability.* Two authors with over five and nine years of experience in scientific workflows independently conduct this evaluation. Below table summarizes the comparative performance of the models.

Although all three models satisfy the correctness and terminology criteria, they differ noticeably across the remaining dimensions. GPT-4o stands out for its clarity and accessibility, producing responses that are especially suitable for foundational learning and novice users. Gemini 2.5 Flash offers the most comprehensive and academically precise answers, making it particularly useful for users with intermediate knowledge or those seeking more formal instructional content. DeepSeek-V3 also generates contextually accurate responses and often enriches them with practical examples; however, its answers are sometimes verbose and less formally structured, which reduces their immediate usability as instructional material without further refinement.

Table: Comparison of LLM Responses across Evaluation Criteria for Core Concepts in Workflows and SWS

| **Criterion** | **GPT-4o** | **Gemini 2.5 Flash** | **DeepSeek-V3** |
|---|---|---|---|
| **Correctness** | ✔ Accurate but basic definitions | ✔ Most precise with academic rigor | ✔ Technically sound with practical examples |
| **Clarity** | ✔ Very clear and straightforward | ✔ Clear but occasionally formal | ⚠ Clear but Verbose |
| **Completeness** | ⚠ Covers basics but lacks depth i.e., omits advanced concepts like validation | ✔ Comprehensive answers with good coverage of key concepts | ✔ Covers most concepts explicitly, often using enumeration |
| **Terminology** | ✔ Uses familiar and readable technical terms. | ✔ Uses formal and precise academic terminology | ✔ Appropriate with real-world SWSs (e.g., Galaxy, KNIME) |
| **Structure** | ✔ Consistent and clean | ✔ Well-organized | ⚠ Often uses lists and informal bullets |
| **Usability** | ✔ Excellent for beginner tutorials | ✔ Good for intermediate audiences | ⚠ Needs cleanup for instructional material |

**Comparative Evaluation of LLM Responses to Galaxy and Nextflow Background Questions:** We next evaluate the responses to the platform-specific background prompts designed to assess the LLMs' contextual understanding of Galaxy and Nextflow . The evaluation results are summarized in the following Table. It reveals distinct strengths and limitations across the three models.

GPT-4o produces accurate, well-structured, and beginner-friendly responses characterized by clear language and logical organization. However, it occasionally omits advanced implementation details, such as containerization strategies and execution environments, which are important for reproducibility and deployment. Gemini 2.5 Flash delivers the most academically rigorous and comprehensive responses, offering precise explanations that cover tool integration, ecosystem capabilities, and platform-specific components. Its slightly formal tone and structured presentation make it particularly well-suited for intermediate users or instructional contexts. DeepSeek-V3 performs well in technical accuracy and real-world applicability, often referencing specific tools and platforms such as ToolShed and WorkflowHub. However, its responses are frequently verbose, rely heavily on bullet-point formatting, and lack the narrative flow necessary for effective instructional use. Overall, Gemini provides the most complete and balanced output, GPT-4o excels in clarity and accessibility for novice users, and DeepSeek-V3 offers valuable practical insight but would benefit from refinement to improve instructional usability.

Comparison of LLM Responses across Evaluation Criteria for Background Questions about Galaxy and Nextflow

| **Criterion** | **GPT-4o** | **Gemini 2.5 Flash** | **DeepSeek-V3** |
|---|---|---|---|
| **Correctness** | Accurate but concise; omits container/cloud details | Most precise with deep tool/system understanding | Correct with real examples from SWSs |
| **Clarity** | Clear and beginner-friendly | Clear with slight formal tone | Verbose and uses list-heavy structure |
| **Completeness** | Covers basics; lacks advanced implementation details | Thorough with ecosystem, portability, and usage depth | Covers broad scope and tool support explicitly |
| **Terminology** | Accessible language for broad users | Formal academic terminology well-used | Includes domain-specific terms and tools (e.g., ToolShed, WorkflowHub) |
| **Structure** | Narrative, well-flowing format | Well-structured with natural coherence | Checklists over narrative, uneven depth |
| **Usability** | Suitable for introductory tutorials | Great for intermediate bioinformatics learners | Needs formatting edits for teaching or documentation |